

Nerve Entrapment

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NERVE ENTRAPMENT IN THE SHOULDER

The incidence of entrapment neuropathies involving the shoulder is largely unknown, but as our understanding of the anatomy and pathology of these conditions grows, earlier and more accurate recognition of these conditions has led to higher rates of treatment. Symptom presentation is most often subtle and insidious in onset, necessitating diagnostic acumen on behalf of the treating physician. Injuries resulting from direct trauma, as with blunt or penetrating mechanisms or from traction across a tethered segment, may present more acutely and be masked by other associated traumatic injuries. A thorough understanding of neurovascular anatomy, common injury mechanisms, and certain at-risk populations should prompt early clinical suspicion. Nerve entrapment is no longer considered simply a diagnosis of exclusion; therefore appropriate advanced imaging and electrodiagnostic examinations are being more readily utilized to confirm the diagnosis. Evidence-based decision-making principles guide operative and nonoperative efforts to return the athlete to competition. This chapter outlines several of the more commonly encountered compressive neuropathies of the shoulder girdle.

SUPRASCAPULAR NERVE PALSY

Anatomy and Biomechanics

The suprascapular nerve arises from the C5 and C6 nerve roots at the upper trunk of the brachial plexus. It passes deep to the trapezius and the omohyoid muscles to enter the supraspinatus fossa through the suprascapular notch beneath the transverse scapular ligament (Fig. 54.1). The suprascapular notch is most commonly U-shaped (48% to 84%) but varies in morphology from flat to enclosed within bone.^{1,2} A study of over 200 scapulas described six types of scapular notches and their incidence as follows: (1) absent (8%), (2) shallow V-shape (31%), (3) U-shaped with parallel margins (48%), (4) deep-V shape (3%), (5) type III with partial ossification of the transverse scapular ligament (6%), (6) complete ossification of the ligament.² Together, the suprascapular notch and the overlying ligament form the suprascapular fossa. The suprascapular notch has been shown to be approximately 4.5 cm from the posterolateral acromion,³ a relationship that becomes important in arthroscopic decompression.

The nerve traverses the deep surface of supraspinatus, innervating it with two motor branches. Sensory branches innervate the glenohumeral and acromioclavicular joints, and as recent

anatomic studies show additional sensory innervation to the coracohumeral ligament, the coracoclavicular ligament, subacromial bursa, and posterior shoulder capsule.⁴ No cutaneous sensory distribution occurs from the suprascapular nerve. Upon reaching the lateral edge of the spine of the scapula, the nerve descends through the spinoglenoid notch, entering the infraspinatus fossa and innervating the infraspinatus.

The spinoglenoid ligament passes from the spine of the scapula to the glenoid neck and posterior shoulder capsule. Its attachment into the posterior capsule results in tightening of the spinoglenoid ligament with cross-body adduction and internal rotation. The ligament is present in 14%⁵ to 100%⁶ of patients. It may appear as a thin fibrous band, known as type I (60%), or a distinct ligament, known as type II (20%), or it may be absent (20%).⁷ The spinoglenoid ligament has been reported to be present more commonly in men (64% to 36%),⁸ although in another study it has been reported to be present in equal proportions in men and women.⁶ The average distance from the supraglenoid tubercle to the nerve at the suprascapular notch is 3 cm.⁹ The distance from the glenoid rim to the spinoglenoid notch is 1.8 to 2.1 cm.^{3,9}

The suprascapular nerve is relatively fixed at its origin in the brachial plexus and at its terminal branches into the infraspinatus, putting it at risk for traction injuries and multiple predictable sites of compression.¹⁰ Nerve contact with the suprascapular ligament is accentuated with depression, retraction, or overhead abduction of the shoulder.² Extremes of scapular motion also place the nerve under tension.^{11–13} Overhead abduction of the shoulder with simultaneous eccentric contraction of the infraspinatus may result in compression of the suprascapular nerve at the spinoglenoid notch.¹⁴ This has been well described in volleyball players.^{14–16} Nerve compression against the lateral margin of the spine of the scapula by supraspinatus and infraspinatus tendons at their point of juncture is also thought to result in nerve injury.¹⁴

“Athletic stress,” especially in throwing or serving, produces a backward and forward rotation of the scapula and suprascapular nerve compression at the suprascapular notch.¹⁷ The nerve is often injured in athletes as it passes around the lateral spine of the scapula and enters the spinoglenoid notch, sparing the supraspinatus.¹⁸ Ganglion cysts are a common cause of compressive injury to the suprascapular nerve. These cysts are thought to result from superior labral tears, which are very common in overhead athletes, with the cyst expanding into the posterior

Abstract

The incidence of entrapment neuropathies involving the shoulder is largely unknown, but as our understanding of the anatomy and pathology of these conditions grows, earlier and more accurate recognition of these conditions has led to higher rates of treatment. Symptom presentation is most often subtle and insidious in onset, necessitating diagnostic acumen on behalf of the treating physician. Injuries resulting from direct trauma, as with blunt or penetrating mechanisms or from traction across a tethered segment, may present more acutely and be masked by other associated traumatic injuries. A thorough understanding of neurovascular anatomy, common injury mechanisms, and certain at-risk populations should prompt early clinical suspicion. Nerve entrapment is no longer considered simply a diagnosis of exclusion; therefore appropriate advanced imaging and electrodiagnostic examinations are being more readily utilized to confirm the diagnosis. Evidence-based decision-making principles guide operative and nonoperative efforts to return the athlete to competition. This chapter outlines several of the more commonly encountered compressive neuropathies of the shoulder girdle.

Keywords

shoulder
nerve compression
decompression
muscle transfers

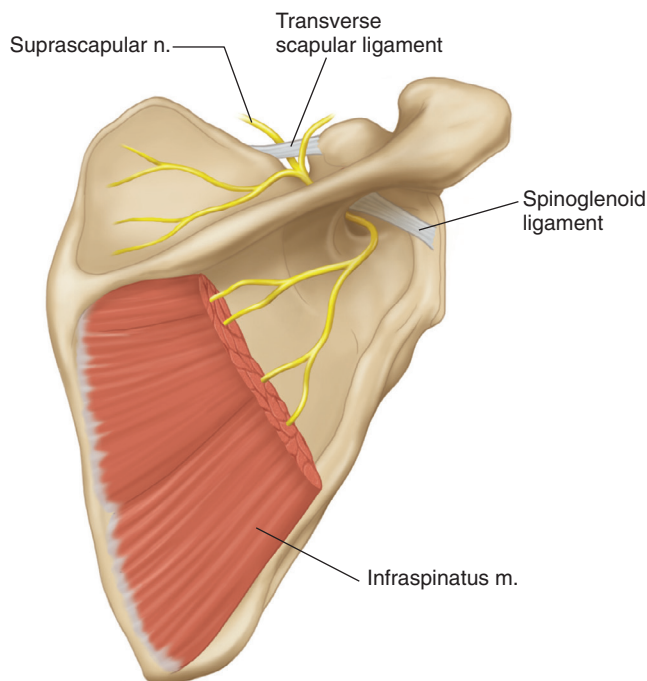


Fig. 54.1 Anatomy of the suprascapular nerve. *n*, Nerve; *m*, muscle. (Modified from Black KP, Lombardo JA. Suprascapular nerve injuries with isolated paralysis of the infraspinatus. *J Sports Med.* 1990;18(3):225–228.)

scapular region, which is devoid of overlying muscle or tendon. Compression of the infraspinatus branch typically occurs as the nerve passes through the spinoglenoid notch (Fig. 54.2),¹⁹ although cysts at the suprascapular notch have been described.²⁰ Other etiologies of neuropathy are more rare but have also been described. Suprascapular nerve palsy has been reported after distal clavicular fractures and resection of the distal clavicle.^{21,22} The nerve is located within 1.5 cm posterior to the clavicle and within 2 to 3 cm of the acromioclavicular joint (Fig. 54.3). Scapular body fractures are also associated with suprascapular nerve palsy.²³ Injury to the axillary or suprascapular artery in overhead athletes can result in microemboli and microvascular infarct of the suprascapular nerve as well.¹⁰ Suprascapular nerve injury has also been reported with acute shoulder dislocation in a cyclist.²⁴

Recent attention has been given to the association of suprascapular neuropathy and retracted rotator cuff tears. A cadaveric study showed increased tension on the first motor branch of the suprascapular nerve with medial retraction of the supraspinatus muscle.²⁵ Clinical studies have suggested that predominant infraspinatus fatty degeneration in the context of a massive rotator cuff tear may be secondary to suprascapular nerve compression at the spinoglenoid notch.²⁶ Repair of the rotator cuff with appropriate tension, however, has been shown to reverse a significant proportion of suprascapular nerve symptoms in the setting of massive tears.²⁷

Clinical Evaluation

Suprascapular nerve injury may present after specific trauma acutely or more commonly chronically. The presentation may be painful range of motion (ROM) or shoulder weakness or

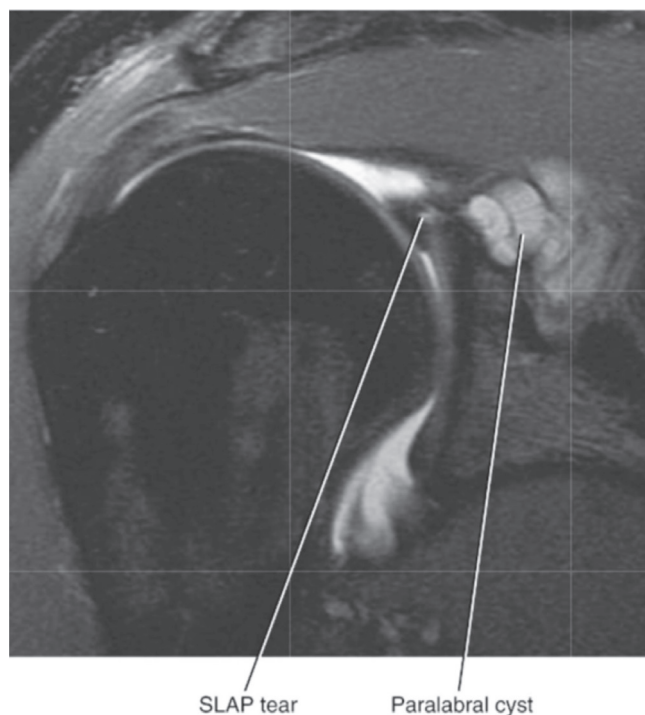


Fig. 54.2 SLAP tear with adjacent paralabral cyst. Coronal T2-weighted image demonstrates a paralabral cyst compressing the suprascapular nerve by mass effect. SLAP, superior labrum anterior to posterior.

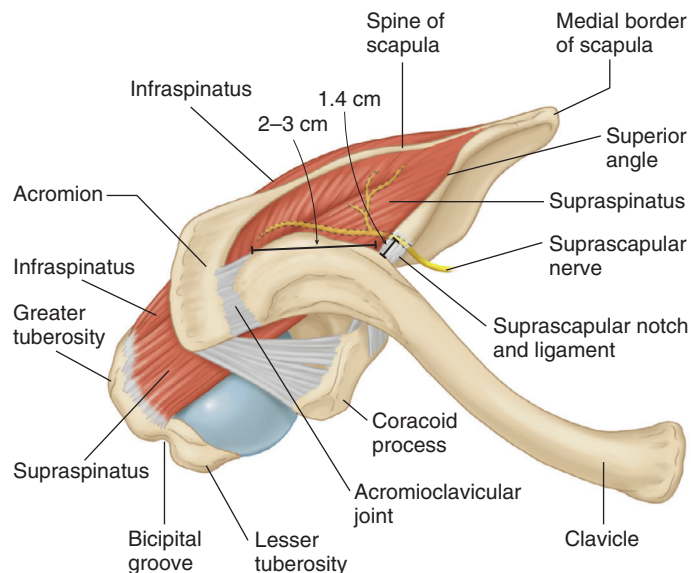


Fig. 54.3 A superior view of the suprascapular nerve showing its proximity to the acromioclavicular joint and posterolateral clavicle. *n*, Nerve.

involve insidious painless muscle atrophy. Athletes who perform repetitive overhead activities—such as baseball players, volleyball players, and swimmers—should be treated with a heightened suspicion for compressive neuropathy.^{10,15,28} During the history-taking portion of the exam, all recent upper-extremity and neck injuries, sports-related or not, should be reviewed. A direct blow or forceful scapular protraction may cause traction on



Fig. 54.4 Rotator cuff atrophy.

the nerve at the root level or kinking at either the suprascapular or spinoglenoid notch.²⁹

The physical exam should begin with general inspection of both shoulders and scapula, concentrating on symmetry of musculature. Scars from trauma or previous surgical incisions should be noted. Active, passive, and resisted ROM is tested, looking for motions that elicit symptoms, the presence of instability, compromised strength, and labral pathology. Tenderness at the suprascapular notch may be elicited with palpation.³⁰ The inconsistent finding of pain often localizes in the posterior shoulder and radiates to the arm; it may be worse with adduction of the shoulder.³⁰ An additional physical exam maneuver, the suprascapular nerve stress test, will worsen compression at the suprascapular notch and produce posterior shoulder pain in patients with compressive pathology.^{31,32} Bilateral symptoms should raise suspicion for cervical pathology, and the exam should always include a c-spine exam.

When the injury occurs at the suprascapular notch, pain and motor weakness of both the supraspinatus and infraspinatus muscles may result. The complaint is most often a gradual onset of vague posterior shoulder discomfort and weakness. Pain is thought to arise from the articular branches to the acromioclavicular and glenohumeral joints.

Compression of the suprascapular nerve by a ganglion at the spinoglenoid notch is a well-known clinical entity.^{33–37} If the lesion is at the spinoglenoid notch, distal to the acromioclavicular and glenohumeral branches, the presentation may be one of painless atrophy of the infraspinatus and external rotation weakness. Posterior shoulder atrophy, especially in the infraspinatus fossa, is an important finding (Fig. 54.4) and should be compared with the contralateral side. Supraspinatus atrophy may be difficult to observe because of the overlying trapezius. Likewise, supraspinatus weakness is not as easily elicited as that of the infraspinatus. Patients with clinical signs suggestive of a labral tear and wasting of the infraspinatus muscle warrant further diagnostic workup, including magnetic resonance imaging (MRI) and electrodiagnostic nerve studies. When the diagnosis is in question, an injection of local anesthetic at the level of the

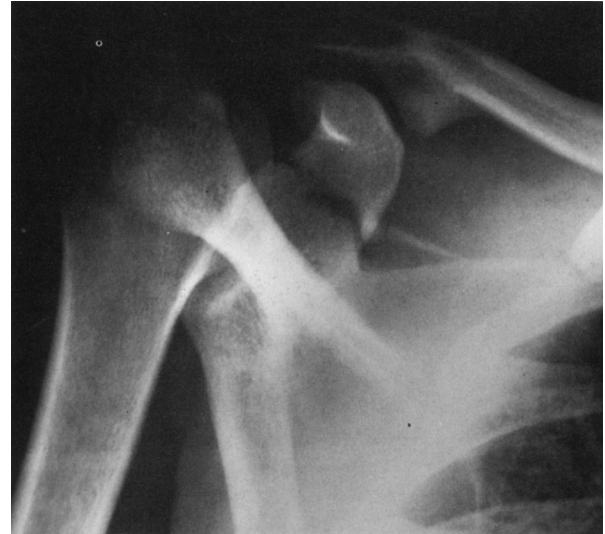


Fig. 54.5 A radiograph of a suprascapular notch fracture with a 30-degree cephalic tilt.

spinoglenoid notch can be helpful to distinguish suprascapular nerve pathology from other causes of shoulder pain.

Diagnostic Studies

Radiographic evaluation should include routine shoulder views and, if clinically indicated, a c-spine series. A 30-degree cephalic tilt radiograph to visualize the suprascapular notch is helpful, especially in patients with fractures (Fig. 54.5).

MRI is useful in the evaluation of patients with suprascapular nerve palsy.³⁸ Acute entrapment may be differentiated from chronic injury on T2-weighted images based on increased signal in the affected supraspinatus and/or infraspinatus muscles. The high signal intensity of affected muscle returns to normal after recovery, as noted on clinical and electrodiagnostic examination.³⁹ Chronic compression appears as typical denervation changes, including decreased bulk and fatty infiltration of the muscles.⁴⁰ Ganglion cysts in the supraspinatus fossa causing compression of the suprascapular nerve can readily be identified on MRI, as can associated pathology such as superior labrum anterior to posterior (SLAP) and rotator cuff tears. Less common causes of suprascapular nerve palsy such as schwannoma⁴¹ and interneural ganglion⁴² have also been identified on MRI.

Ultrasound is also reported to be an effective diagnostic tool for the identification of paralabral cysts and associated rotator cuff tears, although it is very operator-dependent.⁴³ This modality has the added benefit of facilitating guided aspiration of paralabral cysts, with symptomatic improvement in 86% of patients in one series.⁴⁴

Electrodiagnostic evaluation is the gold standard study for suprascapular nerve pathology. Indications include unexplained persistent shoulder pain, atrophy, and weakness without a concomitant rotator cuff tear or MRI evidence of supraspinatus or infraspinatus atrophy without obvious rotator cuff pathology.⁴⁵ A complete test should include both needle electromyography (EMG) of the entire shoulder girdle, including the paraspinal muscles, and nerve conduction velocity (NCV) studies from

Erb's point (2.5 cm superior to the clavicle, representing the convergence of the C5 and C6 nerve roots) to the supraspinatus. Normal conduction latency is 1.7 to 3.7 ms to the supraspinatus and 2.4 to 4.2 ms to the infraspinatus. EMG abnormalities may also be present with brachial neuritis, cervical root compression, and incomplete brachial plexopathies. Conversely, findings of EMG studies may be normal with an obvious clinical suprascapular nerve deficit,^{46,47} confirming the need for the nerve conduction examination. In a study of 79 patients with muscle weakness, EMG and NCV have been shown to have an accuracy of 91% in detecting suprascapular neuropathy.⁴⁸ Overall sensitivity and specificity of electrodiagnostic testing is not known and depends on the etiology of the neuropathy. Compression with ganglia may involve only one of the three or four suprascapular nerve branches to the infraspinatus; therefore EMG recordings are performed at more than one location within the muscle.

Decision-Making Principles

Conservative management with activity modification and physical therapy is the treatment of choice for the majority of patients who present with neuropathy secondary to repetitive overhead activities. Depending on the duration of symptoms and the etiology, a comprehensive physical therapy regimen will likely decrease pain and improve function. However, muscle bulk may be irreversible once significant atrophy has occurred, even if surgically decompressed.⁴⁹ A patient with a chronic condition (i.e., lasting 6 to 12 months) and well-established atrophy that has failed nonoperative measures requires surgical exploration and decompression, as does a patient with suprascapular nerve palsy associated with an acute scapular fracture in the area of the suprascapular notch.²³ Symptomatic patients who have a structural source of compression, such as a ganglion cyst, also benefit from surgical decompression.^{20,33–37}

Spinoglenoid Notch Compression

Suprascapular nerve palsy associated with a spinoglenoid ganglion can be treated with arthroscopic techniques to repair or debride associated labral lesions and decompress the labral cyst.⁵² The cyst may be decompressed arthroscopically through a preexisting labral tear; however, if no labral tear is present, a capsulotomy is performed with an electrocautery device or a shaver. The cyst is visualized with the arthroscope in the posterior portal. A blunt probe is placed in the labral tear or capsulotomy until the characteristic amber-colored cyst fluid is visualized. Decompression is achieved by placing a shaver within the cyst and evacuating the fluid. The cyst wall may be removed with use of the shaver, but care must be taken to avoid iatrogenic injury to the suprascapular nerve. The shaver is pointed at the glenoid neck during removal, and dissection should not extend more than 1 cm medial to the posterior labral attachment to the glenoid.³⁷ Associated labral pathology is then addressed at the same time as the decompression.

If open decompression of the nerve at the spinoglenoid notch is necessary with excision of ganglia, a surgical approach to the posterior glenoid is performed. This approach is begun with a deltoid split over the glenohumeral joint with limited deltoid detachment laterally from the acromion. The superior edge of



Authors' Preferred Technique

Suprascapular Notch Compression

Traditionally open surgical release methods have been utilized to decompress the suprascapular nerve at the suprascapular notch. This was often difficult to visualize open, as the suprascapular nerve is as small as 2 mm and often required detachment of the trapezius muscle for adequate exposure. As our understanding of the anatomy of the area has improved, along with our arthroscopic techniques, multiple all-arthroscopic surgeries have been described and the all-arthroscopic technique has been reported to be a safe and effective alternative to an open approach; this is now considered the standard of care,^{31,50,51} and we prefer to utilize an all-arthroscopic technique at our institution.

The patient is placed in the standard beach-chair position. The all-arthroscopic procedure is as described by Lafosse.^{31,51} Standard posterior and lateral subacromial portals are utilized for camera visualization. An anterolateral portal and suprascapular nerve portal are used for instrumentation. The suprascapular nerve portal is made under direct visualization approximately 7 cm medial to the lateral border of the acromion between the clavicle and the scapular spine and 2 cm medial to a standard superior-medial portal. Once the glenohumeral joint has been visualized through the posterior portal, we enter the subacromial space. The coracoacromial (CA) ligament is identified, exposed, and traced back to its insertion on the coracoid. The coracoclavicular (CC) ligaments are then identified as the dissection is continued posteriorly and medially from the CA insertion. The transverse scapular ligament's lateral border is just medial to the base of the CC ligaments. The suprascapular portal is then made under direct visualization from the lateral portal. A combination of shavers and radiofrequency ablation devices are used to isolate the ligament. Care is taken to avoid the suprascapular artery, which runs superior to the ligament. Another portal, 1.5 cm lateral to the suprascapular nerve portal, is used with a biter to release the transverse scapular ligament. Gentle manipulation of the nerve with a probe should suffice to determine the adequacy of decompression. A burr can be used to open up the notch if there is residual compression after the ligament is released.

the infraspinatus is identified and, at most, the upper half of that tendon is detached, leaving a humeral side stump for repair.⁵³ The size of the exposure needed is based on the MRI position of the ganglion and the size of the patient.

Results

Nonoperative

The overall response to conservative management is good, especially in the absence of a rotator cuff tear or space-occupying lesion. For overhead athletes, rest from sports or other inciting causes may be helpful. The timing of return to activity relies on the judgment of the physician and is based on factors in the course of follow-up, including the extent of the initial paralysis, findings of electrical studies, symptoms, and improvements in the muscle examination with therapy. In persons who have painless infraspinatus muscle palsy without a cyst, function is usually good with nonoperative care. Asymptomatic ganglia without nerve findings may not require treatment.^{14,52,54}

EMG studies have demonstrated that only 30% to 40% of the maximal strength of the infraspinatus is used during overhead throwing; thus in the case of a partial nerve injury, a return to pitching is possible.¹⁸ In a study of asymptomatic volleyball players, it was found that 12 of 96 players had isolated partial infraspinatus paralysis predominantly in the dominant shoulder.

Whereas some players had only electrical abnormalities, others demonstrated muscle atrophy in addition. A 15% to 30% loss of external rotation power was found, although the players retained their ability to play volleyball.¹⁵ The suggested etiology was nerve tension at the spinoglenoid notch when the arm was cocked in maximal external rotation and during follow-through. In a long-term study, subjects with isolated infraspinatus atrophy were reexamined at a mean of 5.5 years. All subjects were still able to play volleyball at a high level with the degree of atrophy unchanged. The incidence of subacromial impingement in this subject population was no higher than that in the general population of volleyball players.¹⁴ Surgical exploration of a well-localized lesion should be performed if 3 to 6 months of conservative management have failed to elicit improvement. Delaying surgery beyond 6 months can lead to irreversible atrophy or incomplete symptom resolution.

Suprascapular Notch Decompression

Most reports show good return of function in selected patients after open surgical decompression.^{55–57} A retrospective review of 42 patients at a mean of 18 months demonstrated measurable improvements in motor strength, with supraspinatus strength improving from grades 0 to 2 to grade 4 in 90% of patients. Infraspinatus strength improved from grades 0 to 2 to grade 3 or better in 32% of patients. Pain symptoms improved almost uniformly.⁵⁷ In recent years arthroscopic suprascapular notch decompression has gained in popularity and become the preferred method of treatment for many orthopaedic sports surgeons.^{31,58} Short- to medium-term follow-up results have been encouraging, restoring patient function and alleviating pain even in the absence of rotator cuff disease.^{31,51,59}

Spinoglenoid Notch Decompression

Several authors have reported cyst resolution and return of nerve function after arthroscopic decompression of the cyst.^{20,33–37,59} Cyst resolution and return of nerve function after repair or débridement of an associated SLAP lesion without attempts at cyst débridement have been reported.⁶⁰ In a large series of patients diagnosed with suprascapular nerve compression, 65 were found to have spinoglenoid notch ganglion cysts associated with glenoid labral tears.³⁶ Patients with the highest degree of satisfaction (97%) were treated with labral repair and open or arthroscopic decompression of the cyst. Lower satisfaction rates were reported by persons treated with isolated labral repair (67%) or needle aspiration of the cyst (64%) and by those treated nonoperatively (53%).

LONG THORACIC NERVE PALSY

Anatomy and Biomechanics

The long thoracic nerve arises from ventral rami of roots of C5, C6, and C7, which branch shortly after they exit from the intervertebral foramina. Branches of C5 and C6 form the upper trunk of the nerve, which pierces through the middle scalene muscle. It then joins the lower trunk from C7, which passes anterior to the middle scalene to form the long thoracic nerve. This pure motor nerve, with a mean length of 30 cm, courses posterior to

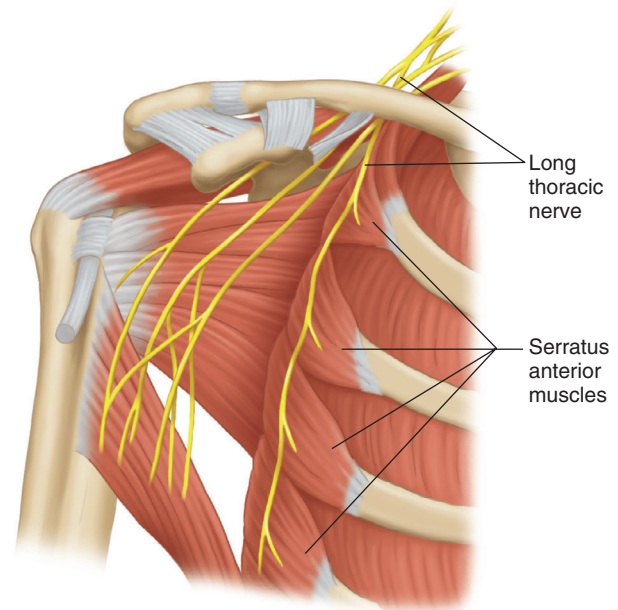


Fig. 54.6 The brachial plexus. (Modified from Haymaker W, Woodhall B. *Peripheral Nerve Injuries*. Philadelphia: WB Saunders; 1956.)

the brachial plexus to perforate the fascia of the proximal serratus anterior (Fig. 54.6).⁶¹ The nerve supplies a single muscle, the serratus anterior, which covers much of the lateral thorax and acts with the trapezius to position the scapula for elevation. It arises from the upper nine ribs and attaches at the deep surface of the scapula along the vertebral border. Innervation of the upper and intermediate portions of the muscle is supplied by the upper division of the long thoracic nerve, which produces shoulder protraction. The lower portion is primarily responsible for scapular stabilization.⁶² These portions typically work together to draw the scapula forward and rotate its inferior angle upward. In patients with serratus anterior muscle paralysis, shoulder motion is severely limited because of the lack of scapular rotation. The serratus anterior also acts as an accessory inspiratory muscle, as is seen in runners who fix their scapulas by holding their thighs to catch their breath after a race.

The most common site of compression is where the nerve passes through the middle scalene muscle or angulates over the second rib.⁶³ A tight fascial band has been identified between the inferior aspect of the brachial plexus and the region of the middle scalene insertion on the first rib.⁶⁴ The long thoracic nerve “bowstrings” over this band with shoulder abduction and external rotation. Medial and upward rotation of the scapula further compress the nerve. Asynchronous motion between the arm and scapula has been implicated as a cause of internal traction injury to the nerve.⁶⁵

Clinical Evaluation

Isolated serratus anterior palsy may result from acute injury, chronic irritation, or brachial neuritis. Long thoracic nerve palsy may also occur with prolonged recumbency or intraoperative stretch during thoracic surgery. Sports and repetitive overhead work have been implicated as a cause of isolated serratus anterior

palsy, with the proposed mechanism being traction injury (either single or repetitive) to the long thoracic nerve.^{24,61,66} In one series the repetitive trauma of tennis and archery was thought to be the cause of the lesion in 5 of 20 patients. Other sports implicated in this type of injury are swimming, basketball, football, golf, gymnastics, ballet, and wrestling.^{67,68} Occupational injuries are also certainly possible, with reports from household activities such as hedge clipping or car washing to construction jobs and shoveling.^{24,69} Traumatic injuries are typically neuropraxias from sudden depression of the shoulder girdle. Proposed traumatic mechanisms include crushing of the nerve between the clavicle and the second rib, tetanic scalenus medius muscle contraction, and nerve stretch with cervical spine flexion or rotation and lateral tilt with ipsilateral arm elevation or backward arm extension.⁶¹ Because the nerve is in a deep location, a direct blow is unlikely to cause isolated palsy. Serratus anterior rupture has been reported in patients with rheumatoid arthritis,²⁹ and injury has been reported as a complication of interscalene blocks administered in the course of regional anesthesia.⁷⁰ All of a patient's past surgical procedures should be reviewed as iatrogenic causes for injury have been described as well, including lymph node biopsies, clavicular resections, and any other procedure performed at the head/neck/shoulder junction.

The long thoracic nerve is often affected by the poorly understood syndrome of brachial neuritis. Parsonage and Turner⁷¹ coined the term *neuralgic amyotrophy* (brachial neuritis), which they found in 136 military personnel, 30 of whom had isolated serratus anterior paralysis. These investigators also noted a right-sided predominance. Significant pain lasting a variable time, from

days to weeks, precedes loss of function in one or more shoulder girdle muscles.⁷² Sensory loss does not exclude the syndrome.

A patient with early thoracic nerve palsy may present with subtle changes in the ability to perform his or her sport, along with decreased active ROM of the shoulder and altered scapulohumeral rhythm. The onset may be painful, as in persons with brachial neuritis, or it may be more subtle, involving difficulties with weight lifting or the recognition of pressure from a chair against the "winging" (protruding) scapula while sitting. Scapular winging may not become evident until several weeks after an acute injury, perhaps because a certain amount of time is required for progressive attenuation of the trapezius to occur.⁶¹

Paralysis of the serratus anterior results in medial winging and poor scapular stabilization, limiting active shoulder elevation to 110 degrees in patients with complete lesions.⁶¹ Scapular winging as a result of long thoracic nerve palsy is characterized by elevation and retraction of the scapula as it moves toward the midline and slightly superior.⁶⁵ The deformity is accentuated by resisted active arm elevation or by performing a push-up maneuver while leaning against a wall (Fig. 54.7). A shoulder protraction test can identify long thoracic nerve injuries affecting the upper trunk.⁶² In this test, the patient is placed in the supine position and asked to forward flex (protract) the shoulder. Ability to protract the shoulder indicates an intact upper trunk of the nerve. Occasionally a Tinel sign over the trajectory of the long thoracic nerve is present.

The appearance of winging with arm elevation due to serratus anterior palsy differs from that of winging due to trapezius palsy. When the serratus anterior muscle does not function, the inferior

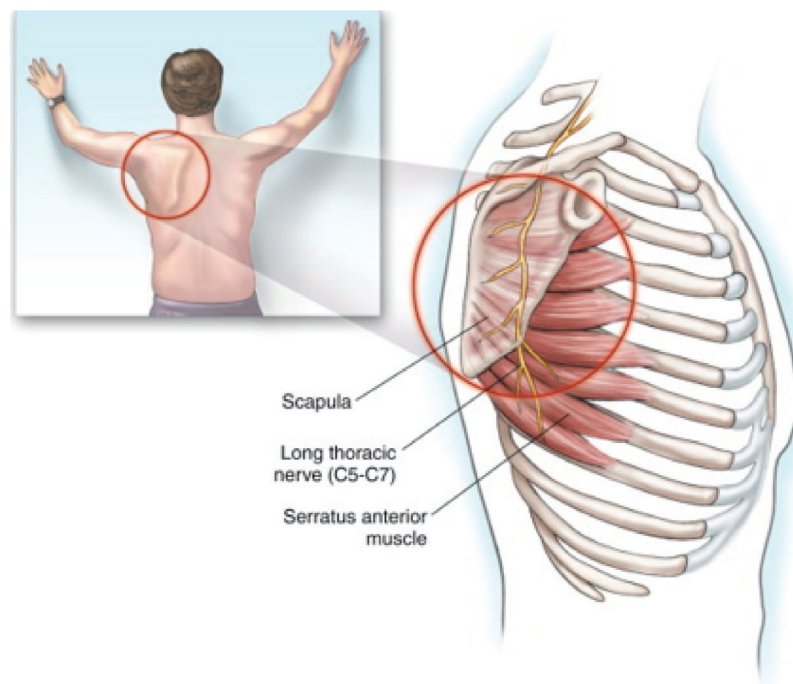


Fig. 54.7 Scapular "winging" is often discovered during weight training as the scapula protrudes with resisted elevation or contacts the flat surface during bench pressing. If weight lifting is thought to be the cause of the problem, participation should cease until nerve function returns. Return to sports by patients with long thoracic nerve palsy depends on the demands placed on the upper extremity by the sport. (From Waldman S. *Atlas of Uncommon Pain Syndromes*. New York: Elsevier; 2014.)

tip of the scapula is pulled medially and posteriorly. With trapezius paralysis, the scapular body is held in position and the medial border merely becomes more prominent, which is a more subtle deformity.

Diagnostic Studies

Plain radiographs of the cervical spine and shoulder may identify contributing arthrosis, fracture malunions, or the presence of accessory ribs and osteochondromas. MRI is generally reserved for patients with shoulder instability or cuff tears,⁷³ although T2-weighted fast-spin echo with fat saturation may demonstrate increased signal intensity consistent with denervation edema.⁷⁴ EMG studies confirm the diagnosis of long thoracic nerve palsy. Conduction studies should be performed from Erb's point to the serratus anterior muscle on the anterolateral chest wall. Repeat nerve studies at 3- to 6-month intervals are helpful to monitor improvement.⁷³

Decision-Making Principles

Surgery may be indicated for patients with long thoracic nerve palsy who remain symptomatic and for whom electrodiagnostic studies show no improvement despite 1 to 2 years of nonoperative therapy. Athletes must be counseled that no surgical intervention will provide a reliable return to a competitive sport that requires overhead strength.⁷⁵

Results

Nonoperative

The outcome of acute traction injuries is typically good with conservative management, and spontaneous functional improvement has been well described.^{2,73} In a series of 17 patients presenting with various etiologies at a mean of 3 months who were treated with a scapula winger's brace, investigators reported an increase of one motor grade with the brace applied.⁸³ Six patients demonstrated brace-free recovery to near premonitory function by final follow-up. The investigators hypothesized that the mechanism was proprioceptive feedback preventing muscle overuse or attenuation and transfer of contralateral shoulder mechanics. The prognosis for spontaneous recovery from Parsonage-Turner brachial neuritis is good; 36% of patients diagnosed with this syndrome recover by the end of the first year and 75% recover by the end of the second year.⁸⁴ Further improvement may occur beyond 2 years.²⁴ Recurrent long thoracic nerve palsy is rare.⁶¹

Pectoralis Major Transfer

No surgical series consisting exclusively of athletes has been reported and functional outcomes and return to work statistics are mixed. A systematic review⁸² suggested that interpositional grafts have a higher risk of recurrent medial scapular winging, possibly due to graft attenuation over time. Indirect transfers also had a mean of 12 degrees less postoperative ROM, but functional scores were still equivalent compared with direct pectoralis transfer. Only 46% of patients were able to return to their previous level of work, and although winging may have been resolved, pain because of underlying neuropathy or altered scapulohumeral kinetics often persisted. Forward shoulder flexion was reliably restored in a series of patients who underwent a split tendon transfer at



Authors' Preferred Technique

Long Thoracic Nerve Palsy

Cessation of the activity that is suspected of inciting the injury is important. Shoulder braces cannot begin to normalize rotational balance of the scapula due to the force couple between the serratus anterior and the trapezius. However, in the case of severe serratus winging, braces may prevent progressive attenuation of the trapezius muscle.

Surgical treatment is infrequently indicated. Although pectoralis minor transfer^{76,77} to the lateral inferior scapula for dynamic support has been reported, transfer of the pectoralis major sternal head, with or without augmentation, to the inferior border of the scapula is the currently favored reconstruction owed to its similar line of pull, excursion, and cross-sectional area as compared with the serratus anterior.^{73,78,79} Indirect transfers with interpositional grafts have been described with equivalent functional results as a direct transfer. The advantage of the interpositional graft is that it avoids overtensioning, which would restrict postoperative ROM⁸⁰ and reduce the risk of traction neuropathia to the medial and lateral pectoral nerves.⁸¹ There is considerable controversy in the literature regarding the most effective surgical technique. A recent study failed to show a consensus among polled shoulder and elbow surgeons regarding the use of the entire tendon or a split tendon technique, direct versus indirect transfers, or regarding graft type.⁸²

The patient is placed in the lateral position. A 5-cm incision is made in the inferior deltopectoral groove. The shoulder is abducted to accentuate the interval between the sternal and clavicular heads of the pectoralis major. The sternal head is released sharply from its humeral insertion and bluntly separated from the clavicular head. Care is taken to work medially against the chest wall to avoid the neurovascular structures. Sutures are placed in the tendon to maintain proper orientation during shuttling through the second incision. A second incision is made over the inferior angle of the scapula. The infraspinatus, subscapularis, and serratus anterior are subperiosteally dissected to expose the inferomedial portion of the scapula. A long curved clamp is used to develop a tunnel in the scapulothoracic interval, medial to the coracoid process and conjoined tendon. The transferred pectoral tendon is then secured to the scapula through drill holes with the scapula held in a reduced position.⁷⁹ Postoperative immobilization in a sling or scapulothoracic orthosis is emphasized for 6 weeks. Gentle gravity-assisted pendulum exercises and isometrics are allowed during this time. ROM of the shoulder girdle begins 6 weeks after surgery. Return to noncontact athletic activities begins at 12 weeks, but lifting of more than 20 lb is delayed until 6 months after surgery.

a mean of 6 years after their palsy commenced.⁷⁹ In a series of 11 consecutive patients who were followed up for a mean of 41 months, one patient demonstrated full recurrence of scapular winging.⁸⁵ In a critical analysis of results of 15 patients at a mean of 64 months,⁸⁶ all patients demonstrated decreased ROM in the operative shoulder. Seven of 15 had good to excellent results, and 13 returned to their previous employment. Objective scapular winging was measured with the shoulder abducted 90 degrees while holding as much weight as possible. Only one patient had more than mild winging, but no patient lifted more than 25 lb.

SPINAL ACCESSORY NERVE PALSY

Anatomy and Biomechanics

The spinal accessory nerve (cranial nerve XI) exits the jugular foramen at the base of the skull. It penetrates and innervates the upper third of the sternocleidomastoid muscle before crossing the posterior triangle of the neck. It is here that this nerve is

superficial and vulnerable to injury. The nerve enters the trapezius and is the predominant motor nerve to that muscle. Root fibers from C3 and C4 also innervate the trapezius and may blend with the accessory nerve, although their contribution may be only proprioceptive.⁸⁷ The accessory nerve is a small purely motor nerve, measuring only 1 to 3 mm in diameter.⁸⁸

Scapular stabilization and elevation result from the balance between the trapezius and serratus anterior. The upper trapezius elevates and tilts the scapula, raising the lateral shoulder and assisting in arm elevation, respectively. The lower trapezius works with the rhomboids to retract the scapula and balance the pull of the serratus anterior.

Clinical Evaluation

Spinal accessory nerve injuries are rare in sports. They typically occur as a result of a direct blow, as with a hockey stick or in a traction injury with a cross-face maneuver in wrestling.⁸⁹ Stretch injury resulting from distal upper extremity distraction and

contralateral head rotation has been reported.⁹⁰ Iatrogenic injuries, again, are a common cause of spinal accessory nerve injuries; therefore a patient's surgical history should be reviewed closely.

The patient can present with a sagging shoulder at rest and incomplete arm elevation with loss of strength with shoulder shrug (Fig. 54.8). Trapezius atrophy can also be seen. The symptoms may be quite severe as a result of muscle spasm and brachial plexus traction neuritis or may occur as a more subtle ache, stiffness, and weakness, especially with overhead activity. Examination shows a drooping of the shoulder and a deepening of the supraclavicular fossa after trapezius atrophy has occurred. Winging of the scapula occurs with resisted arm elevation and active external rotation against resistance.⁹¹ The superior angle moves laterally compared with the inferior angle and gives the classic "lateral winging" appearance. The levator scapulae is palpable and is seen as a band of muscle in the neck; rhomboid contraction is also palpable on attempted scapular adduction. The lateral winging is accentuated in abduction but will disappear when



Fig. 54.8 Starting top left and moving clockwise: (A) Posterior view of a patient with a left spinal accessory nerve palsy attempting a shoulder shrug. (B) Note the drooping left shoulder at rest and trapezius atrophy. (C and D) Anterior view of patient attempting shoulder shrug and at rest with a left spinal accessory nerve palsy.

the arm is flexed forward, unlike the medial winging seen with serratus anterior palsy. Lateral winging is typically less obvious and is often less disabling with trapezius palsy than with serratus anterior palsy. However, shoulder function in an athlete with accessory nerve palsy is often inadequate for competition.

Diagnostic Studies

EMG is used to provide the definitive diagnosis, but its role in determining prognosis for recovery has been questioned.⁹² Furthermore, low-amplitude compound muscle action potentials and voluntary motor unit potentials may be present in the context of severe spinal accessory nerve injury, warranting surgical exploration.⁹³

Decision-Making Principles

Functional results with closed injuries that are treated conservatively are mixed.^{65,92,94,95} Spinal accessory nerve palsy does not seem to respond as well to conservative therapy as serratus anterior palsy. Neuropraxic lesions can often be monitored for up to 1 year before considering surgical intervention.⁹⁴ If there is complete injury to the nerve, often as it runs in the posterior cervical triangle, and it is diagnosed within 1 year, microsurgical nerve reconstruction should be considered. Symptoms persisting longer than a year typically require the Eden-Lange muscle transfer procedure. Clinical indications for exploration or reconstruction include a symptomatic patient with upper extremity drooping, aching, numbness, and incomplete active arm elevation. Adjacent scapular muscles cannot substitute for a paralyzed trapezius with muscle strengthening alone.



Authors' Preferred Technique

Spinal Accessory Nerve Palsy

The current reconstruction of choice, called the Eden-Lange procedure, involves transferring the levator scapulae, rhomboid minor, and rhomboid major laterally to more functionally advantageous positions to substitute for the upper, middle, and lower trapezius.⁹⁵ With the patient in the lateral position, the head of the table is elevated 15 degrees. A longitudinal incision is made midway between the spinous processes and the medial border of the scapula along its entire course. The trapezius is identified and released close to the scapular border, taking care not to injure the underlying rhomboids. The levator scapulae, rhomboid minor, and rhomboid major are separated from each other and then detached from the scapula with a thin strip of insertional bone. These muscles are then elevated with care to avoid injuring the dorsal scapular nerve and transverse cervical artery. Each muscle is tagged individually with a suture, and the infraspinatus is then elevated from its fossa. Six longitudinal drill holes are placed through the scapula 1.5 to 2 cm apart, with the most superior hole 4 to 5 cm lateral to the medial border of the scapula, just inferior to the scapular spine. With a large needle, heavy nylon sutures are passed, taking care not to penetrate the chest wall. The rhomboid minor is secured to the two most superior drill holes, and the rhomboid major is secured to the remaining four drill holes. The infraspinatus muscle is then imbricated over the repair. A second skin incision is made over the scapular spine 3 cm medial to the posterior tip of the acromion, extending medially 4 cm. The trapezius, deltoid, and supraspinatus muscles are then elevated from the scapular spine so that three drill holes can be made through the spine. The levator scapulae is then passed through a tunnel in the trapezius in line with its upper fibers and secured to the scapular spine. After skin closure and application of a dressing, the shoulder is placed in an abduction brace.

Results

In a relatively large series of 27 patients treated for trapezius muscle palsy, 20 underwent neurolysis or repair of the spinal accessory nerve at a mean of 7 months after injury. The remaining 7 patients underwent lateral muscle transfers at a mean of 20 months after the onset of symptoms. Of the 20 patients who underwent nerve surgery, 16 had good or excellent results, with an average follow-up shoulder abduction of approximately 125 degrees. Of 7 patients who underwent muscle transfer, 4 had good or excellent results. Predictors of poor results were patient age greater than 50 years and nerve lesions resulting from radical neck dissection, penetrating injury, or spontaneous spinal accessory nerve palsy.⁹⁶ In a series of 7 patients who were followed for a mean of 39 months, 5 had excellent results as demonstrated by full forward flexion, minimal scapular winging and shoulder droop, and improved pain. The remaining 2 patients, who had each undergone a two-stage reconstruction, had residual pain and weakness.⁹⁵

AXILLARY NERVE PALSY

Anatomy and Biomechanics

The axillary nerve arises from the posterior cord of the brachial plexus with contributions from the C5 and C6 nerve roots. The nerve passes close to the coracoid process before coursing posteriorly just inferior to the shoulder joint capsule. It then enters the quadrilateral space, joined by the posterior humeral circumflex artery, and is bound by the humerus laterally, the long head of the triceps medially, the teres minor and infraspinatus superiorly, and the teres major inferiorly (see Fig. 54.8). The axillary nerve divides into anterior and posterior branches, innervating anterior and posterior portions of the deltoid muscle, respectively. A branch from the posterior division also supplies the teres minor muscle and skin overlying the deltoid insertion. Points of relative fixation at the posterior cord, quadrilateral space, and deltoid muscle render it vulnerable to injury.⁹⁷ The axillary nerve is vulnerable to traction injury after shoulder dislocation⁹⁸ and as a result of direct blunt trauma.⁶⁵ Also, considering the close proximity of the axillary nerve to the inferior glenohumeral capsule as it passes from anterior to posterior, it is at risk for compression from an inferior humeral osteophyte from glenohumeral arthritis.⁹⁹ Most commonly, quadrilateral (or quadrangular) space syndrome is caused by chronic compression of the posterior humeral circumflex artery and axillary nerve within the quadrilateral space. Fibrotic bands along the inferior edge of teres minor and within the space have been implicated,^{100,101} although any space-occupying lesion (e.g., ganglion, aneurysm) or traumatic disruption of the normal anatomy can cause compression.

Clinical Evaluation

Quadrilateral space syndrome is a chronic compressive neuropathy that typically presents in athletes who throw; it is the most common cause of axillary neuropathy.¹⁰² The patient often presents with an insidious onset of abduction weakness, decreased sensation over the deltoid insertion, and eventually

deltoid atrophy and glenohumeral subluxation.⁶⁵ Symptoms are aggravated by forward flexion and/or abduction with external rotation of the humerus.¹⁰⁰ Resisted shoulder internal rotation in this provocative position also exacerbates symptoms of a dull ache.¹⁰³ Discrete point tenderness is noted on palpation posteriorly over the quadrilateral space and a Tinel sign over the quadrilateral space may be elicited.¹⁰⁰ Symptoms are attributed to compression of the axillary nerve, not to arterial occlusion.¹⁰¹ Although quadrilateral space syndrome is the most common cause of compressive axillary neuropathy, it is certainly not the sole cause of axillary palsy; thus a full history must be taken, focusing on previous injuries and surgeries to the shoulder/neck region, and the duration and quality of symptoms. A full musculoskeletal physical exam should be performed, including aspects of a upper extremity neurologic and vascular exam.

Diagnostic Studies

MRI may demonstrate cysts in the quadrilateral space, inferior labral tears, or fatty atrophy of the teres minor muscle.¹⁰⁴ Muscle edema in the teres minor and/or deltoid may be demonstrated in subacute cases.¹⁰⁵ Electrodiagnostic studies may demonstrate increased latencies on nerve conduction velocities and denervation patterns on EMG of the teres minor muscle.^{74,106} Positive subclavian arteriography will demonstrate occlusion of the posterior humeral circumflex artery with the shoulder in abduction and external rotation.¹⁰⁰ An arteriogram may be performed but is not necessary because it suggests associated symptomatic nerve compression only indirectly.¹⁰⁶

Decision-Making Principles

Conservative measures have been successful in up to 90% of patients.¹⁰⁰ Nonoperative treatment consists of cessation of throwing and management of symptoms. Rehabilitation focuses on shoulder and trunk flexibility, strength, and proper throwing mechanics.¹⁰² In the absence of a space-occupying mass or identifiable lesion, conservative treatment is indicated.¹⁰⁷



Authors' Preferred Technique

Axillary Nerve Compression in the Quadrilateral Space

Quadrilateral space decompression is performed with the patient in a lateral position. A 3.5-cm longitudinal incision is made over the posterior shoulder at the point of maximal tenderness. The posterior deltoid is elevated superolaterally to expose the fat within the quadrilateral space. Fibrous adhesions are carefully dissected while palpating the axillary nerve and accompanying posterior humeral circumflex artery along the humeral neck. The arm is placed in maximal abduction and external rotation to ensure unrestricted nerve gliding and a palpable arterial pulse as a finger is easily passed through the quadrilateral space and into the axilla.

The shoulder is placed in a sling for postoperative comfort, and the patient is instructed in the immediate use of gravity-assisted pendulum exercises. Active ROM and gentle strengthening exercises are initiated on postoperative day 10. Full abduction with external rotation is avoided until 4 weeks after surgery. A sport-specific therapy program is initiated at 6 weeks.

Results

Nonoperative

Several case reports discuss the nonoperative management of athletes who perform repetitive overhead movements. Two volleyball players demonstrated improved motor strength and resolution of sensory disturbance after abstaining from competitive sports (for 6 months and 1 year, respectively).¹⁶ A competitive baseball pitcher altered his throwing mechanics to a three-quarter overhead motion and successfully returned to competition.¹⁰³

Operative

The surgical solution for axillary nerve palsy depends, as always, on the etiology. A well-described surgical procedure for brachial plexus or axillary nerve injuries in patients with loss of active shoulder abduction is transfer of the triceps branch of the radial nerve to the anterior branch of the axillary nerve. A combined case series showed that the nerve transfer was successful in the majority of patients, increasing shoulder abduction strength and active ROM.¹⁰⁸ In the original series of patients presenting with quadrilateral space syndrome, 18 underwent posterior decompression. Eight experienced complete relief, eight showed improvement, and two did not show any improvement.¹⁰⁰ In a subsequent review of five patients who experienced a traumatic event (two traction injuries and three injuries that occurred as a result of a fall), it was reported that pain symptoms improved and sensory deficits over the lateral arm resolved in all patients. Three patients had persistent limitation of active motion as a result of their associated shoulder joint injuries.¹⁰⁶ In a series confined to athletes who performed repetitive overhead movements, four patients presented with disabling posterior shoulder pain, despite normal findings on electrodiagnostic examination.¹⁰⁹ Upon decompression of the quadrilateral space, three patients were found to have fibrous bands constricting the axillary nerve, and one was found to have constricting venous dilation. All four patients returned to full pain-free activity without restriction by 12 weeks after surgery.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Freehill MT, Shi LL, Thompson JD, et al. Suprascapular neuropathy: diagnosis and management. *Phys Sportsmed*. 2012;40(1):72–83. doi:10.3810/psm.2012.02.1953.

Level of Evidence:

V

Summary:

The authors report a thorough review of the etiology, diagnostic studies of choice, and recent updates in surgical treatment.

Citation:

Lafosse L, Piper K, Lanz U. Arthroscopic suprascapular nerve release: indications and technique. *J Shoulder Elbow Surg*. 2011;20(2 suppl):S9–S13. doi:10.1016/j.jse.2010.12.003.

Level of Evidence:

VII

Summary:

The authors present their own indications and technique for arthroscopic suprascapular nerve release at both the spinoglenoid and suprascapular notch.

Citation:

Chalmers PN, Saltzman BM, Feldheim TE, et al. A comprehensive analysis of pectoralis major transfer for long thoracic nerve palsy. *J Shoulder Elbow Surg.* 2015;24(7):1028–1035. doi:10.1016/j.jse.2014.12.014.

Level of Evidence:

V

Summary:

The authors present the results of a survey of shoulder and elbow surgeons to record a recent snapshot in time their technical preference in pectoralis major transfers for long thoracic nerve palsies. They also provided a systemic review of the outcomes of the differences in technique.

Citation:

Bigliani LU, Perez-Sanz JR, Wolfe IN. Treatment of trapezius paralysis. *J Bone Joint Surg Am.* 1985;67(6):871–877.

Level of Evidence:

IV

Summary:

The authors present a series of patients who sustained an injury to the spinal accessory nerve resulting in paralysis of the trapezius muscle. The rationales for surgical decision-making, surgical technique, and functional outcomes are presented.

Citation:

McAdams TR, Dillingham MF. Surgical decompression of the quadrilateral space in overhead athletes. *Am J Sports Med.* 2008;36(3):528–532. doi:10.1177/0363546507309675.

Level of Evidence:

IV

Summary:

The authors present a series of four athletes who perform repetitive overhead movements and had axillary nerve entrapment within the quadrilateral space, resulting in deltoid muscle paralysis. The authors present a modified surgical technique and report functional outcomes.

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